

Mini Project 1 Part A

Submission: The required documentation uploaded to Blackboard for this part of Mini Project 1 should contain the following:

- A word document containing your source code (This can be copied and pasted in from your CCS file). Make sure your code is thoroughly commented to explain what is happening in your code. Marks will be lost for poor commenting or just using my comments that were in the skeleton code.
- Photos of your solution if you got it working.
- Screen shots of relevant register contents e.g. ADC10MEM and various stages of your program, e.g. for Q1 a screen shot of the register when the light is on and then one when the light is off. You can also include screen shots of any register you think is important to show.

Note: You should put all exercise solutions in a single word document.

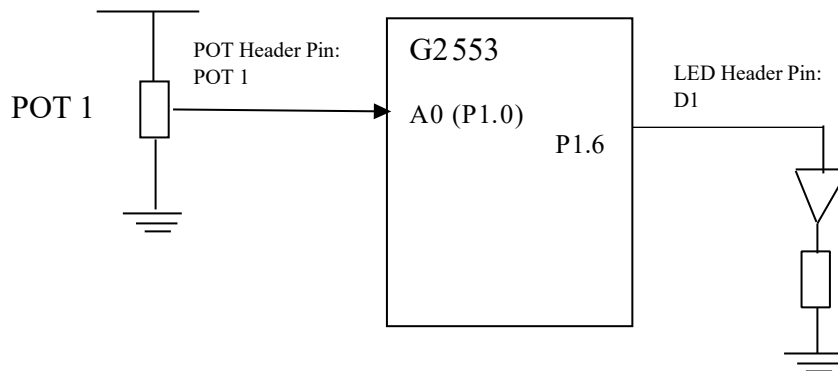
Q1.

An external analog voltage (potentiometer) is connected to the external analog A0 input of the G2553 (analog channel 0) – refer to the G2553 pin-out given in notes.

To get you started use the code provided in the MP_Part1A1.txt file in the 1st weeks lab folder in Blackboard. This shows an outline of how the code should be structured and includes some bits of code to start you off. All you need to do is set the registers to the correct values where you see question marks.

Your final code should:

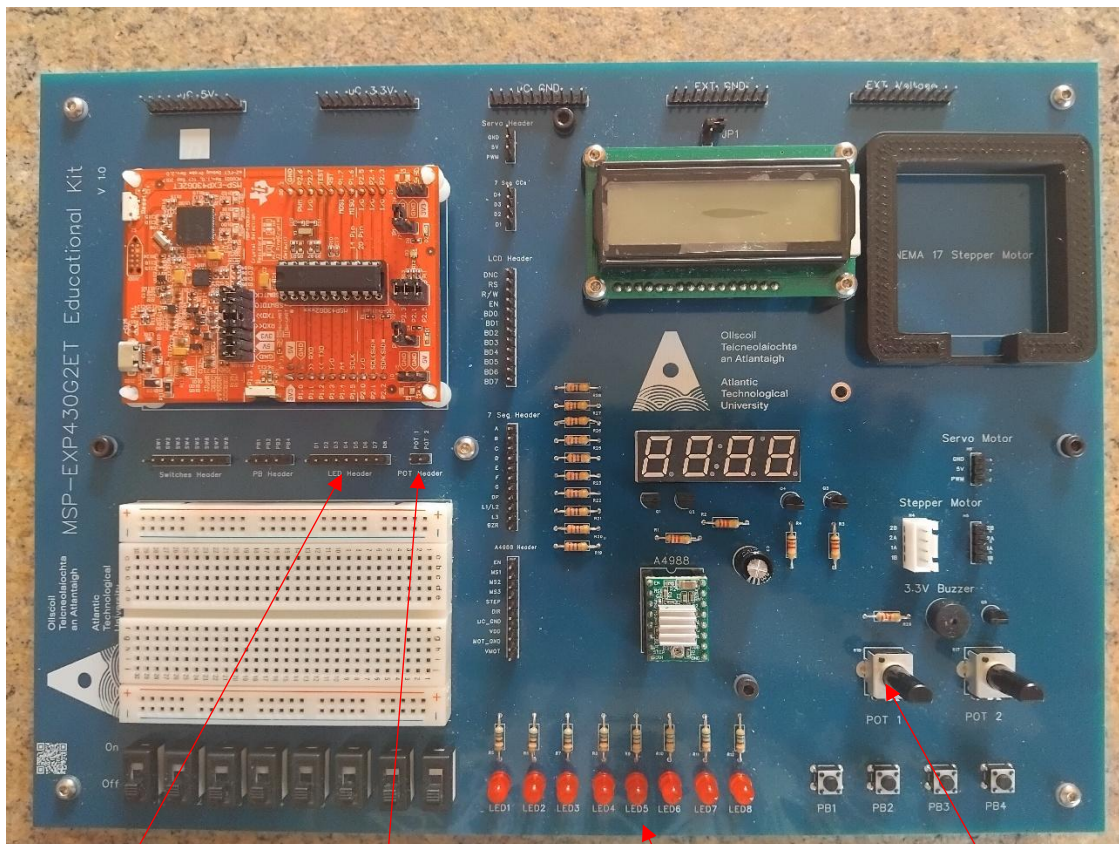
- Configures the Port Pins correctly
- Enables analog input on A0. This is done using the ADC10AE0 register – refer to notes
- selects this channel (A0) as the input to the ADC – see ADC10CTL1 register for details
- uses a reference equal to the G2553 supply voltage i.e. 3.3V. This means we can apply an analog signal in the range 0-3.3V to the A0 pin for conversion.
- Performs continuous (every second, approximately) Analog to Digital conversions on the A0 channel and lights an LED (connected on P1.6) whenever the input voltage exceeds 1.65V (i.e whenever the ADC conversion result is greater than 512)
- Test your code by varying the pot and observing the behaviour of the LED



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For this circuit we will use the onboard LEDs and POT1 of the MSP-EXP430G2ET Educational Kit (See Below)

Note: Use Female-Female Jumper Wires to connect between the MSP430 Pins and the Hardware Header pins on the educational Kit

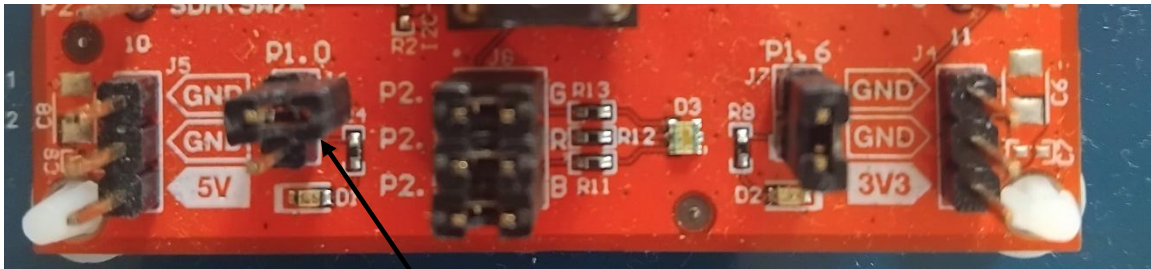


LED Header

POT Header

LEDs

POT 1



Note: Remember to remove the jumper on the P1.0 port pin, on Launchpad, before testing. It is the jumper at the bottom left of the board. **Please do not lose the Jumper!!!**

Q2.

- An external analog voltage (potentiometer) is connected to the external analog A1 input of the G2553 (analog channel 1) – refer to the G2553 pin-out given in notes. A Red LED is connected on P1.6 on Launchpad board and a Green LED on P1.0 of Launchpad board
- Note: Make sure Jumpers 1.0 and P1.6 are both connected on the MSP430 board and that the Educational Kit LED1 is disconnected

Write code which

- Performs **continuous** Analog to Digital conversions on the A1 channel (monitor V_{A1} , the analog input voltage on A1) and lights the Launchpad LEDs (connected on P1.0 and P1.6). as follows;
 - If $0 \leq V_{A1} < 1V$ then don't light any LEDs
 - If $1V \leq V_{A1} < 2V$ then light the GREEN LED only
 - If $2V \leq V_{A1} < 3V$ then light the RED LED only
 - If $V_{A1} > 3V$ then light both LEDs
- Test your code by varying the pot and observing the behaviour of the LED

Q3 Part A

Write a C function, to which the channel number on which a conversion is to be performed, is passed, as an integer. The function should configure the ADC (3.3 V_{ref} etc., etc.) for the required channel, perform a conversion and return the integer answer.

Refer to notes on functions from previous modules, as necessary.

Q3 Part B

Using the Function developed in Q3 Part A to configure Ports A4 and A5, write a program which compares two analog input voltages on analog input channels A4 and A5 and behaves as follows:

- If the ADC results for A4 and A5 are within 10 units of each other (i.e. voltages are almost equal) the RED and GREEN LED is on (P1.6, P1.0)
- If the voltage on A4 is greater than that on A5, only the Green LED should be on
- If the voltage on A4 is less than that on A5, only RED LED should be on..
- Neither input voltage will exceed 3.3V
- Sample the voltages once every ¼ of a second.
- Connect the Educational Kit onboard POTS 1 and 2 to the A4 and A5 inputs and test your code.